

Hexagonal Finite Volume Method - Convergence Analysis

Rigorous numerical validation of hexagonal finite volume methods for integration over Lipschitz domains

This repository contains the complete implementation and validation suite for the paper:

“Rigorous Error Analysis of Hexagonal Finite Volume Methods on Lipschitz Domains”

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Overview

The hexagonal finite volume method (FVM) achieves $\mathbf{O(n^2)}$ convergence for numerical integration over bounded Lipschitz domains, with explicit error constant:

$$C_1 = (5 \cdot A \cdot M \cdot d^2) / 24$$

where:

- A = domain area
- $M = \|f\|_{\{W^2, \infty\}}$ (Sobolev norm bound)
- d = domain diameter

Why Hexagons?

Hexagons provide the **optimal spatial covering** among regular polygons (packing density $\sqrt{3}/2 \approx 0.9069$), a property observed in nature (bee honeycombs) and proved mathematically by Hales' honeycomb conjecture. This translates to superior numerical properties: six-fold rotational symmetry yields isotropic discretization error.

Installation

```
# Clone the repository
git clone https://github.com/abdullah-dabdoub/hexagonal-fvm-convergence.git
cd hexagonal-fvm-convergence

# Install dependencies
pip install -r requirements.txt
```

Requirements

- Python 3.8
- NumPy 1.20
- SciPy 1.7
- Matplotlib 3.4
- Pandas 1.3
- Shapely 1.8

Usage

Quick Start

Run the complete validation suite:

```
python hexagonal_fvm.py
```

This will:

- 1. Compute analytical reference integrals for all test cases
- 2. Run hexagonal FVM at refinement levels $n = 16, 32, 64, 128, 256$
- 3. Fit convergence orders and compare with theoretical bounds
- 4. Generate publication-quality figures (PNG + PDF)
- 5. Save summary CSV to `results/convergence_summary.csv`

Output Structure

```
results/
├─ convergence_summary.csv      # Numerical results table
├─ figures/
│  ├─ conv_Disk_f1.png         # Disk + affine/bilinear
│  ├─ conv_Disk_f2.png         # Disk + oscillatory
│  ├─ conv_Disk_f3.png         # Disk + quadratic
│  ├─ conv_Disk_f4.png         # Disk + near-singular
│  ├─ conv_Square_f1.png       # Square + affine/bilinear
│  ├─ conv_Square_f3.png       # Square + quadratic
│  ├─ conv_LShape_f1.png       # L-Shape + affine/bilinear
│  ├─ conv_LShape_f3.png       # L-Shape + quadratic
│  └─ ... (PDF versions)
```

Test Suite

Domains

Domain	Type	Area (A)	Perimeter (K)	Diameter (d)
Unit Disk	Smooth		2	2
Unit Square	Piecewise-smooth	1	4	$\sqrt{2}$
L-Shape	Non-convex	1.5	6	$2\sqrt{2}$

Test Functions

Function	Formula	$M = \ f\ _{\{W^2,\infty\}}$	Description
f	$1 + xy$	0	Affine/bilinear (exact on symmetric cells)
f	$\sin(x)\cos(y)$	$2 \approx 9.87$	Oscillatory
f	$1 + x^2 + y^2$	2	Quadratic (primary convergence test)

Table 2 – continued

Function	Formula	$M = \ f\ _{\{W^2,\infty\}}$	Description
f	$\sqrt{((x-0.5)^2 + (y-0.5)^2 + 0.01)}$	1	Near-singular

Expected Results

All non-exact test cases confirm $\mathcal{O}(n^2)$ convergence with observed constants within 23–49% of the theoretical bound C , indicating the bound is conservative but correctly ordered.

Key Features

- ✓ **Analytical reference integrals** — no polygon approximation error
- ✓ **Machine-precision detection** — exact integration cases labeled correctly
- ✓ **Theoretical bound overlay** — compare observed vs. predicted errors
- ✓ **Publication-quality figures** — 300 DPI PNG + vector PDF output
- ✓ **Reproducible** — fixed random seed, deterministic algorithms

Mathematical Background

Core Theorem

Theorem (Convergence). Under standard regularity assumptions, the hexagonal FVM with midpoint quadrature satisfies:

$|R_H(f, a_n) - I[f]| \leq C_1 \cdot n^{-2} + \mathcal{O}(n^{-3})$

with explicit constant $C = 5AMd^2/24$.

Proof Ingredients

- Steiner’s Formula** for Minkowski neighborhood volume
- Boundary cell enumeration** via geometric measure theory
- Taylor expansion** with vanishing first-order term (centroid property)
- Optimality** proof for single-point evaluation schemes

Citation

If you use this code in your research, please cite:

```
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  title={Rigorous Error Analysis of Hexagonal Finite Volume Methods on Lipschitz Domains},
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```

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